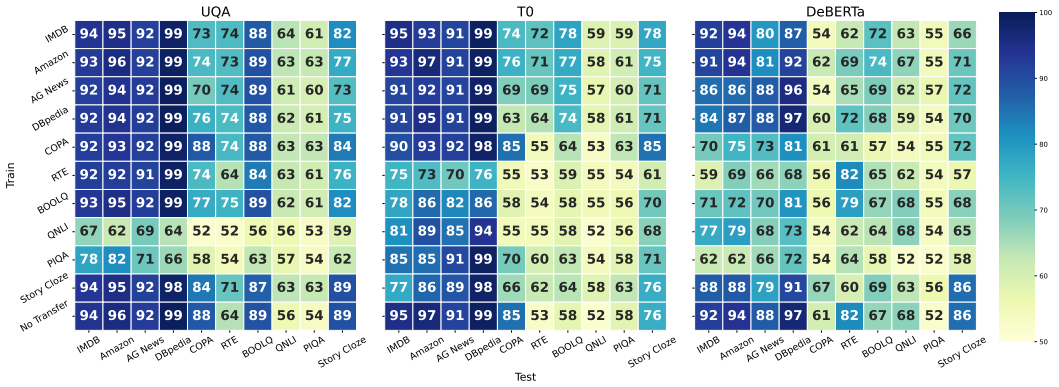

Figure 2: Transfer Performance using CCS on UnifiedQA, T0 and DeBERTa. The y-axis corresponds to the training dataset, and the x-axis corresponds to the test dataset. The final row ("No Transfer") corresponds to training and testing on the same dataset, which is the same as the diagonal. On most datasets, CCS transfers well to other datasets (relative to no transfer), including to different tasks with completely different labels. In some cases transfer even outperforms the no-transfer setting. See Appendix E for results with other models.

middle of the network outperform hidden states at the end of the network when using CCS (see Figure 10). This is especially true for UnifiedQA on misleading prefixes; we find that using the encoder hidden states is robust to misleading prefixes (Section 3.2.2), but that accuracy using the decoder hidden states drops from 81.0% to 73.5%, a similar amount to zero-shot accuracy. This suggests that compared to the later layers of a model, intermediate layers are more robust and less correlated with the model outputs, and that CCS can take advantage of this.

Finally, if CCS were just recovering knowledge in the model outputs, we would only expect it to work in cases where model outputs are informative. However, we show in Appendix C that CCS still works with masked language models when their outputs are uninformative: when we don't [MASK] any input tokens, and when we prompt models so that the labels used to construct contrast pairs appear in the *middle* of a prompt rather than at the end. These results show that CCS can sometimes recover latent knowledge in a model that is distinct from—and more useful than—what the model outputs.

3.3.3 Truth is a Salient Feature

From the results we have presented so far, it is possible that the direction learned by CCS is difficult to find and requires using a large amount of unsupervised data. We provide evidence against this possibility by showing that finding such a direction can both (1) often be done with only a small amount of data, and can also (2) often be done by essentially taking the top principal component of a slightly modified representation space.

CCS doesn't require much data. We now evaluate how well CCS performs with different amounts of data. Whereas before we trained CCS using the full training set and all prompts, here we use limited data and a single prompt. Specifically, we train using only k unlabeled contrast pairs, using the single prompt for each model and dataset that achieves the highest zero-shot accuracy. We still test on all prompts for each dataset. We resample k points 32 times for each of $k = 1, 2, 4, \cdots$, and take the average accuracy across those 32 samples. Finally, we plot the average such accuracy across all datasets and prompts for several models in Figure 3.

We find that while CCS benefits from more data, it can often do well with very limited data. In fact, it can sometimes even do well with only a single contrast pair, though we find high variance across individual datasets; see Appendix D for more details. This suggests that the strong performance of CCS does not primarily come from using a large amount of unsupervised data, and indicates that the direction learned by CCS may be relatively easy to find.

Contrastive Representation Clustering. We now show that directions correlated with the truth may be "salient" in a different way: by showing that we can also find such directions using either (1) PCA or (2) clustering. Specifically, suppose we construct contrast pairs (x_i^+, x_i^-) as before. Intuitively, these two examples are qualitatively almost identical except that one is true and the other

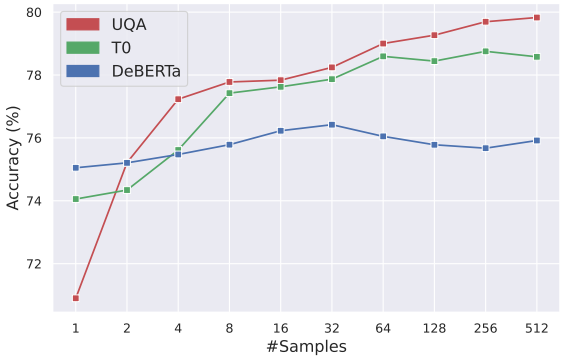

Figure 3: Accuracy when we train CCS on k samples for different values of k (each time averaged across 32 trials). We use the single prompt with the highest zero-shot accuracy for each dataset and model. While CCS benefits from more examples, it can often work well with limited data.

is false, so the main difference between the representations $\tilde{\phi}(x_i^+)$ and $\tilde{\phi}(x_i^-)$ should relate to truth. Consequently, we can take the differences in (normalized) hidden states, $\{\tilde{\phi}(x_i^+) - \tilde{\phi}(x_i^-)\}_{i=1}^n$, and cluster them. We call this method Contrastive Representation Clustering (CRC). Clustering can be achieved by taking the top principal component (TPC) and thresholding at 0, or by doing a "bimodal salience search" (BSS) to find a direction that looks bimodal; see Appendix G.4 for further details.

We compare CCS and these two variants of Contrastive Representation Clustering in Table 2, using the same setting as in Table 1. While CCS performs best, all methods attain high accuracy and are competitive with zero-shot performance. This indicates both that (1) representations of truth often lie in a high-variance direction in the contrastive representation space $\{\tilde{\phi}(x_i^+) - \tilde{\phi}(x_i^-)\}_{i=1}^n$, and also that (2) true and false examples are often well-clustered in this same contrastive space. This strengthens the idea that representations of truth may be salient features inside models that are relatively easy to find. This may help explain why CCS can perform well without using any supervision, and how it can do so even with only a limited amount of unlabeled data.

4 Related Work

Zero-Shot Prompting. Since the release of GPT-3 (Brown et al., 2020), one of the main paradigms for eliciting what models know has been zero-shot prompting (Liu et al., 2022; Beltagy et al., 2022). Zero-shot exploits how language models are trained to predict diverse data from the internet, which incidentally includes tasks such as question-answering. If prompted appropriately, this can be used to solve various useful tasks with reasonable performance (Brown et al., 2020). However, these models are trained to imitate human-generated data, which bounds the quality of their outputs.

Many methods improve upon vanilla zero-shot prompting (Liu et al., 2022; Zhao et al., 2021; Lu et al., 2022; Wei et al., 2022b; Min et al., 2022a). While our goal is not to improve zero-shot performance, some of the ideas underlying these methods are similar to CCS. Particularly relevant are methods that also leverage unsupervised consistency properties, such as Jung et al. (2022); Zhou et al. (2022). However, these methods still bootstrap from language model outputs trained via imitation learning, which limits their applicability to our main goals.

Method	RoBERTa	DeBERTa	GPT-J	T5	UQA	T0*	Mean*
Calibrated 0-shot	64.3(6.2)	76.3(6.0)	56.0(5.2)	58.8(6.1)	80.4(7.1)	90.5(2.7)	67.2(6.1)
CCS	62.1(4.1)	**78.5(3.8)**	61.7(2.5)	**71.5(3.0)**	**82.1(2.7)**	77.6(3.3)	**71.2(3.2)**
CRC (TPC)	**65.7(4.9)**	77.0(4.9)	60.8(3.2)	68.3(4.6)	78.8(3.0)	65.3(6.8)	70.1(4.1)
CRC (BSS)	63.6(5.5)	77.9(4.9)	**61.9(2.3)**	67.4(2.2)	80.0(3.4)	79.0(5.1)	70.2(3.7)

Table 2: We compare CCS to two variants of Contrastive Representation Clustering: TPC, which clusters by projecting onto the top principal component, and BSS, which clusters by finding a direction that looks bimodal. We show accuracy and standard deviation of each model averaged across all prompts and datasets, in the same setting as Table 1. We find that CCS generally performs the best, but that all methods are competitive with zero-shot.

To illustrate this, imagine we train reinforcement learning agents to play a game such as Diplomacy (FAIR et al., 2022), in which players have incentives to lie to each other. Then those agents may learn to lie in a way that is difficult to detect, but they may still internally represent whether they are lying. Because their outputs would be deliberately misleading, standard zero-shot methods may be very unreliable. In contrast, techniques like CCS may still be able to detect whether those models are lying by finding representations of truth in their activations that contradict their outputs.

Truthfulness. There has been increasing interest in making language models truthful (Evans et al., 2021; Lin et al., 2022). One aspect of truthfulness that has received substantial attention is factuality (Thorne et al., 2018; Maynez et al., 2020). For instance, many techniques aim to improve the factuality of models by augmenting them with retrieval methods (Nakano et al., 2021; Menick et al., 2022), which allows them to cite their sources. In contrast, we focus on truthfulness more generally, which also includes procedural knowledge such as reasoning or natural language inference tasks.

An approach to making language models truthful in this more general setting is to finetune them using either human demonstrations (Khashabi et al., 2020; Sanh et al., 2021; Zhong et al., 2021; Wei et al., 2022a) or reinforcement learning from human feedback (Christiano et al., 2017; Stiennon et al., 2020; Askell et al., 2021; Bai et al., 2022; Ouyang et al., 2022). These techniques have been widely successful at improving performance, but unlike our method they rely on being able to provide ground truth labels, which can be intractable in many settings.

Some work has aimed to go beyond the direct supervision humans can provide by augmenting supervision with AI systems (Christiano et al., 2018; Irving et al., 2018; Leike et al., 2018; Perez et al., 2022). This may expand the range of applications we can supervise, but many of these proposals remain theoretical, and it is unclear just how far these techniques can generalize. Christiano et al. (2022) reframes this issue by posing the problem of Eliciting Latent Knowledge (ELK). Like our problem statement, ELK is about eliciting knowledge from models even in cases where humans cannot evaluate that knowledge. However, ELK frames this as a worst-case theoretical problem, while we frame this as an empirical problem that we can make progress on using current models.

5 Discussion

5.1 Limitations and Future Work

Our work has a number of limitations. First, CCS relies on the existence of a direction in activation space that separates true and false inputs well, in the sense that a supervised probe on the activations would be able to attain high accuracy (if it hypothetically had access to the ground truth labels). This requires that a model is both *capable* of evaluating the truth of a given input, and also that the model *actively evaluates the truth* of that input. It is not clear when these conditions hold precisely.

Second, we did not evaluate our method on setups involving active "lying" or "deception" (Kenton et al., 2021; Evans et al., 2021) by models, as we aren't aware of existing evaluation setups for this setting. If future work develops such a setup, a good stress test would be to apply CCS to do "lie detection" in that setting. This may require modifications or extensions to the method, such as more explicitly ensuring that it recovers the truth of an input rather than what the model says.

There are also various straightforward improvements to our method that one could explore. This includes adding additional consistency constraints, improving its reliability, calibrating its probabilities, generalizing it beyond the yes-no question-answering setting, generalizing it to cases where answers aren't clear-cut, and closing the remaining gap between CCS and the logistic regression ceiling.

5.2 Conclusion

As language models become more capable, they will be increasingly used as components in larger AI systems trained with reinforcement learning. As this occurs, falsehoods arising from misaligned training objectives may become more common, severe, and difficult to detect. In principle, models may even develop instrumental incentives to lie: for example, if a human evaluator would disapprove of bad model behavior, models may learn to lie about their behavior to achieve higher reward. If so, those models would be optimizing against human evaluators, making it precarious to rely on those evaluators for assessing the truth of what models say. Because unsupervised methods for eliciting answers circumvent this issue, they may still work even in this scenario. We show that it is possible to make progress on such methods today; the empirical success of our approach suggests that unsupervised methods are both a tractable and underexplored research direction.